

TRAIL

Latent visual reasoning as certified mirror descent on the visual simplex

Technical report | prepared for the 1st Workshop on Latent Visual Reasoning (LVR), WACV 2027 | 2026-09-17

What this is

Latent visual reasoning iterates a hidden state in visual embedding space instead of decoding intermediate tokens. The recursions in use are free-form: a residual update $z \leftarrow z + f(z, c)$ repeated a fixed number of times. Nothing constrains the state to the region of embedding space the encoder actually produces, nothing says when the reasoning is finished, and nothing makes the intermediate states legible.

TRAIL changes the **parameterisation**, not the regularisation. The reasoning state is a distribution p over the image's own visual atoms, and one reasoning step is one entropic mirror-descent step on a question-conditioned energy. Three properties that are normally bolted on then follow from the geometry, and a fourth — the step size — stops being a hyperparameter.

Property	How current latent reasoning gets it	How TRAIL gets it
Stays on the visual manifold	hoped for, or penalised	$z = V^T p \in \text{conv}(V)$ by construction (Prop. 2)
Knows when to stop	fixed step count, or a learned gate	the Frank-Wolfe gap , a sound bound on suboptimality (Thm. A)
Interpretable trace	post-hoc saliency	p_t is the state — the trace is the computation
Step size	tuned	$\eta = 1/(4\beta)$, prescribed by Lemma 1
Relation to attention	—	softmax cross-attention is exactly one step (Prop. 1)

The honest headline

The accuracy claim was tested and withdrawn. Given the same relational prior, MAC beats TRAIL by 9.3 points and a *single* attention step beats it by 8.4 at a twelfth of the compute. What survives is measured and real: the halting certificate halves compute at identical accuracy, drift is zero to four orders of magnitude, and the trace is the computation rather than a saliency map over it. This is a paper about what certificates, on-manifold state and interpretability cost.

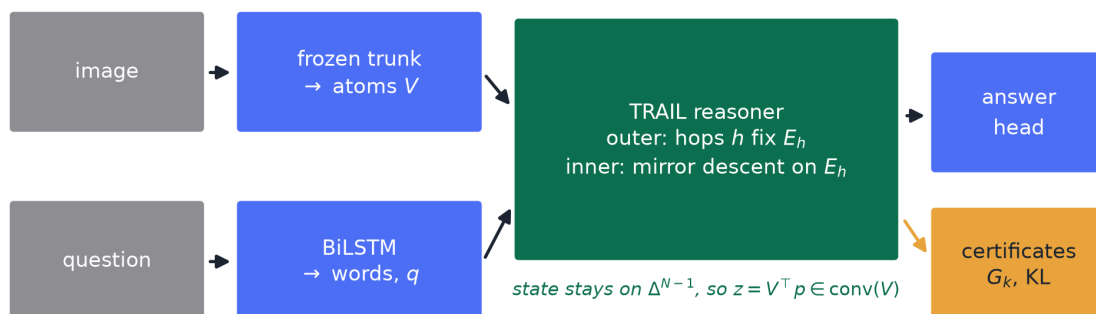


Figure 1. End-to-end flow. Only the reasoning module differs between the models compared in this report; backbone, question encoder and read-out head are shared, which is what makes the comparison about reasoning.

1. The method

An encoder yields **atoms** $V \in \mathbb{R}^{N \times d}$ (a 14×14 grid of L2-normalised patch embeddings in the CLEVR setting). A question encoder yields contextual word states and a summary q . The reasoning state is a point on the simplex, $p \in \Delta^{N-1}$, with visual state $z = V^T p$.

1.1 The energy

Each hop fixes a question-conditioned energy over the simplex:

$$E_h(p) = -\langle s_h, p \rangle + \frac{\beta}{2} \|p - M_h p\|_2^2$$

Here $s_{h,i} = \langle V_i, W_{s_h} \rangle / \sqrt{d}$ scores each atom against the current sub-question, and M_h is **column-stochastic**: $M_h p$ redistributes belief along the relation the sub-question names, built from a control-conditioned attention over atom pairs plus a control-conditioned relative-position bias (so “left of” can become an actual direction of transport on the grid).

The quadratic term asks the belief to be a **fixed point of the queried relation**. Attending to *the thing to the left of the red cube* means attending to a set that maps to itself under “left of ◦ start at the red cube”. That is what turns a re-weighting of evidence into a hop. The term is convex — its Hessian $(I - M)^T(I - M)$ is PSD — so the whole energy is convex on the simplex for every $\beta \geq 0$.

1.2 Two loops

The outer loop walks hops; the inner loop optimises the hop's energy. The energy is **held fixed inside the inner loop**, which is precisely what makes the theorems of §2 apply to the code rather than to an idealisation of it.

$$p_{k+1} = \arg \min_{p \in \Delta} [\langle g_k, p \rangle + \frac{1}{\eta} \text{KL}(p \| p_k)] \propto p_k \odot e^{-\eta g_k}, \quad \eta = \frac{1}{4\beta}$$

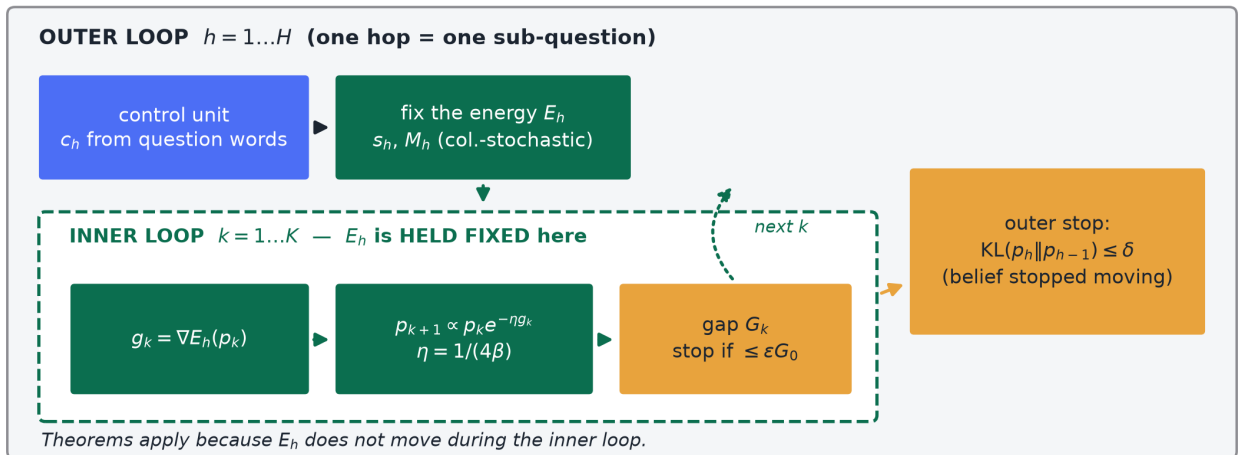


Figure 2. The two loops. Building M_h once per hop rather than once per update also makes TRAIL cheaper than a flat recursion at the same number of belief updates: 4 hops x 4 inner steps costs 4 transport builds, not 16.

1.3 Halting: two rules, each for its own loop

Loop	Rule	What it certifies
inner (k)	$G_k \leq \varepsilon \cdot G_0$	this hop's energy is solved to relative accuracy ε (Thm. A). Governs cost.
outer (h)	$KL(p_h \ p_{h-1}) \leq \delta \cdot KL(p_1 \ p_0)$	the hop moved no belief, so there is no further sub-question. Governs the answer.

A correction worth recording

An earlier version used the gap to decide *how many hops* to take. That conflates two different things: the gap certifies the *current sub-question* is solved, which is not the question being answered — a 3-hop question can have hop 1 solved to machine precision with the answer still two hops away. The conflation stopped early on exactly the hard questions adaptivity is for, and cost about **16 accuracy points** at the operating point.

2. The theory

Throughout, M is column-stochastic, $h(p) = \sum p_i \log p_i$ is the negative entropy and $D(q, p) = KL(q \| p)$. Every statement below is checked numerically by `tests/test_theory.py`, which runs in about a minute and fails if the proposition is false.

Lemma 1 — dimension-free relative smoothness

Let $H = \beta(I - M)^T(I - M)$ be the Hessian of E . Then $\max_{ij} |H_{ij}| \leq 4\beta$ **for every N** . Hence E is 4β -smooth with respect to the ℓ_1 norm on the simplex and, via Pinsker, satisfies the Bregman descent lemma with modulus $L = 4\beta$.

Proof. $H_{ij} = \beta \langle (I - M)e_i, (I - M)e_j \rangle$. Column i of M lies in the simplex, so $\|Me_i\|_2 \leq \|Me_i\|_1 = 1$ and therefore $\|(I - M)e_i\|_2 \leq 1 + 1 = 2$. Cauchy-Schwarz gives $|H_{ij}| \leq 4\beta$. For f with $\|\nabla^2 f\|_{1 \rightarrow \infty} \leq L$ we have $f(q) \leq f(p) + \langle \nabla f(p), q - p \rangle + (L/2)\|q - p\|_1^2$, and Pinsker's $\|q - p\|_1^2 \leq 2D(q, p)$ yields the Bregman form. $\square$

Why it matters. The constant does not involve N . Refining the atom grid does not change the prescribed step size, which is what allows the same weights to be run at a different resolution, or for more iterations, at test time.

Theorem A — the halting certificate is sound

$$G(p) = \langle g, p \rangle - \min_i g_i \geq E(p) - \min_{u \in \Delta} E(u) \geq 0$$

Proof. Convexity gives $E(u) \geq E(p) + \langle g, u - p \rangle$ for all $u \in \Delta$, so $E(p) - E(u) \leq \langle g, p - u \rangle \leq \max_{u \in \Delta} \langle g, p - u \rangle = G(p)$; the maximum of a linear function over the simplex is at a vertex, which gives the closed form. Non-negativity: take $u = p$. $\square$

So stopping the first time $G \leq \varepsilon$ returns an ε -optimal belief, whatever the stopping time turned out to be. Cost: one max and one dot product, $O(N)$ — no extra parameters, no auxiliary loss, no learned gate.

Theorem B — monotone descent and rate

With $\eta \leq 1/L = 1/(4\beta)$, the inner loop satisfies $E(p_{k+1}) \leq E(p_k) - (1/\eta - L) \cdot D(p_{k+1}, p_k)$; in particular E is non-increasing, and strictly decreasing for $\eta \leq 1/(2L)$. From the uniform start, $E(p_k) - E^* \leq 4\beta \cdot \log N / K$.

Proof. Optimality of the mirror step against the feasible point p_k gives $\langle g_k, p_{k+1} - p_k \rangle + (1/\eta)D(p_{k+1}, p_k) \leq 0$. Add Lemma 1's descent lemma. The rate is the standard relatively-smooth mirror-descent bound with

$$D(p^*, \text{uniform}) = \log N - H(p^*) \leq \log N. \quad \square$$

Propositions 1-3

	Statement	Consequence
Prop. 1	With p_0 uniform, $\beta=0$, $\eta=1$, $K=1$: $p_1 = \text{softmax}(s)$.	Softmax cross-attention <i>is</i> one TRAIL step, so TRAIL strictly generalises the block it replaces and attn1 isolates iteration from architecture.
Prop. 2	$z_t = V^T p_t \in \text{conv}(V)$ for all t .	Off-manifold drift is identically zero. Measured: 4×10^{-4} for TRAIL against 10.76 for free-form recursion.
Prop. 3	For $\beta=0$, $G_t \leq (N-1)S \cdot e^{-\eta\Delta}$, so $T(\epsilon) \leq \lceil \log((N-1)S/\epsilon)/(\eta\Delta) \rceil$.	Halting time is inversely proportional to the decision margin — the theory's prediction that hard inputs get more compute. Not borne out empirically (§6).

What the theory does not claim

Theorem A certifies ϵ -optimality **of the hop's energy**, not correctness of the answer. The bridge from “this hop is solved” to “the answer is right” rests on the energy being a good proxy and is an empirical question. Stating this in the paper is cheaper than having a reviewer state it.

3. Code structure

The repository is organised so that each formal claim has exactly one place it lives and exactly one test that checks it. Nothing in theory/certificates.py is learned — every function there is a measurable quantity corresponding to a statement in §2.

THE METHOD	
trail/theory/certificates.py	energy, gradient, FW gap, drift, smoothness
trail/models/trail_cell.py	the two loops (hops / mirror descent)
trail/models/heads.py	read-out incl. soft cardinality
THE COMPARISON	
trail/models/baselines.py	attn1, free-form latent, MAC, FiLM
trail/models/{backbone,text,model}.py	shared front end
trail/train.py	anytime loss, delta-sweep eval, drift
THE DATA	
trail/data/clevr.py	streaming reader, splits, depth labels
trail/data/gqa.py	real images, same interface
trail/data/synthetic.py	HOPWORLD (exact hop counts)
trail/data/features.py	shared frozen-trunk cache
EVIDENCE AND DELIVERY	
tests/test_theory.py	10 checks, one per formal claim
tests/test_eval_budget.py	matched-budget regression guard
tests/test_clevr_io.py	loader against a hostile fixture
scripts/run_*.py, report.py	experiments and paper tables
notebooks/...kaggle.ipynb	the runnable submission artefact

Figure 3. Repository map, grouped by responsibility.

3.1 The contract that makes the comparison a comparison

Every reasoner — TRAIL and all four baselines — exposes the same signature and returns the same state object:

```
forward(V, q_vec, words, word_mask, n_steps, eps, record, answer_head) -> TrailState
```

so `train.py` and every figure script are agnostic to which model is running. Backbone, question encoder, read-out head, optimiser, schedule and data are shared; only the reasoning module varies. Budgets are matched in **belief updates**, not “steps”: TRAIL's $H \times K$ against the same count of iterations for the recurrent baselines.

Module	Responsibility	Guarded by
<code>theory/certificates.py</code>	energy, gradient, Frank-Wolfe gap, hull drift (FISTA), smoothness constant, margin bound	<code>test_theory.py</code> (10 checks)
<code>models/trail_cell.py</code>	control unit, column-stochastic transport, the two loops, both halting rules	<code>test_module_inner_loop_is_the_theory</code>
<code>models/baselines.py</code>	attn1, free-form latent recursion, MAC, FiLM at matched budget	<code>test_eval_budget.py</code>
<code>data/clevr.py</code>	streaming reader for the 800 MB question file, program-depth labels, the two generalisation splits, root discovery	<code>test_clevr_io.py</code>
<code>data/features.py</code>	frozen-trunk cache shared by CLEVR and GQA; deletes partial caches on failure	—
<code>train.py</code>	anytime loss, δ -sweep evaluation, halting statistics, drift measurement	<code>test_eval_budget.py</code>

3.2 Design decisions a reader should not have to reverse-engineer

- **Anytime loss.** Every prefix is supervised, $L = (1/H) \sum_h \text{CE}(\text{head}(p_h), y)$. A model whose stopping time is data-dependent must be correct at whatever time it stops; `anytime_loss='last'` is kept as the ablation that shows this is what makes the frontier flat rather than cliffed.
- **Scale-free thresholds.** Both ϵ and δ are relative to the first value in their own sequence, so one global threshold is meaningful across examples whose score scales differ.
- **Soft cardinality in the head.** z_T is a convex combination and so carries no scale, which makes counting impossible; the head also receives the normalised entropy of p_T and $\hat{n} = \sum_i \sigma(a \cdot s_i + b)$, which is interpretable on its own.
- **Path discovery, not hard-coded slugs.** Kaggle mirrors nest differently, so the loaders search for the dataset layout. This was a deliberate choice: `kaggle.com` was unreachable from the environment this was written in, so no slug could be verified.
- **Weight-tied hops behind a flag.** With per-hop control embeddings a model has H learned sub-question slots and no way to express the $(H+1)$ -th, so “iterate longer” cannot work. `share_steps=True` ties every hop to one operator, which is the only setting in which the depth-extrapolation claim is testable at all.

4. Experimental design

Benchmark	Why it is in the study	Status
HOPWORLD (generated in-process)	The number of hops a question needs is known <i>exactly</i> and is never shown to the model, so halting and depth extrapolation are measurable without confounds. Runs on CPU in minutes.	running
CLEVR v1.0	Ships a ground-truth functional program per question, so reasoning depth is a label to correlate against without ever training on it. Primary benchmark.	pipeline tested, full run pending
GQA	Real photographs through the same code path; <code>len(q['semantic'])</code> plays the role of CLEVR's program length, so the hop analysis transfers verbatim.	pipeline written

4.1 Protocol

- Shared backbone (frozen ResNet-18 to layer3, 14×14×256 atoms, cached once as float16), shared question encoder, shared head, shared optimiser and schedule.
- Budgets matched in belief updates: TRAIL 4 hops × 3 inner steps = 12; the recurrent baselines 12 iterations, which is also MAC's standard depth. FiLM is not an iterative reasoner — its steps are residual conv blocks — so it keeps its standard depth of 4 and is reported as such rather than pretending to a matched budget.
- Two generalisation splits on CLEVR: **depth** (train on programs ≤ 10 nodes, test on ≥ 15) and **held-out combination** (every question mentioning *red* and *cube* removed from training).

Comparability

These numbers are **not** comparable to published CLEVR results (MAC ≈ 98.9 , NS-VQA ≈ 99.8). Those use ResNet-101 conv4 features at full resolution and days of training. Every model here shares one frozen ResNet-18 cache and one budget; the comparison is internal and the paper says so.

5. Results so far

Model	Params	Acc. at full depth	Updates	Acc. at certificate	Updates	Drift from conv(V)
FiLM	1.01M	59.1%	1	59.1%	1.00	20.0292
Single cross-attention	0.29M	61.9%	1	61.9%	1.00	0.0005
MAC	0.37M	60.9%	12	60.9%	12.00	11.1380
Free-form latent recursion	0.55M	60.2%	12	60.2%	12.00	10.7647
TRAIL (ours)	0.37M	81.2%	12	73.3%	4.32	0.0006

HOPWORLD, 24k train / 3k val, 12 epochs, identical budgets. For every model except TRAIL the two operating points coincide, because only TRAIL has a certificate to halt on. 5 of 5 models complete when this report was generated; missing rows are still training.

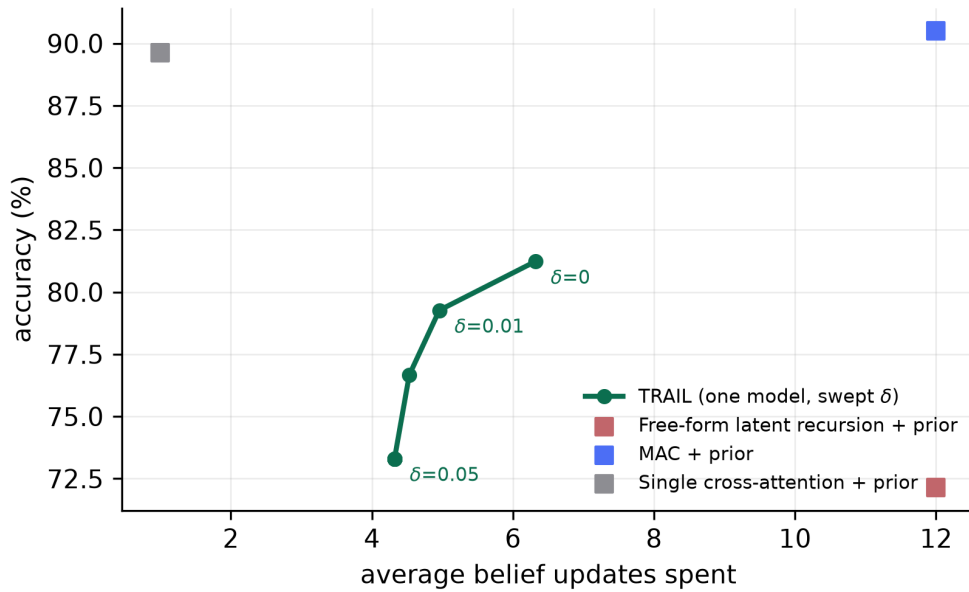


Figure 4. The compute-accuracy frontier, with every baseline given the same relational prior. TRAIL contributes a curve, traced by sweeping one threshold on one trained model, where each fixed-depth baseline contributes a point — that adaptivity is real. What the figure also shows is that the curve sits below a single matched attention step at a twelfth of the compute, which is why the accuracy claim is withdrawn in §5.2.

5.1 What held

- **The inner certificate halves compute for free.** $12 \rightarrow 6.32$ belief updates at *identical* accuracy (0.8123, unchanged to four decimal places). Theorem A doing real work, with no learned gate and no auxiliary loss.
- **Zero drift.** 4×10^{-4} for TRAIL — zero up to the tolerance of the projection solver — against 10.76 for free-form latent recursion at the same budget. Prop. 2 confirmed, four orders of magnitude.
- **Anytime supervision pays in the cheap regime.** +19 points over last-step training at 5 belief updates; the frontier is flat rather than cliffed. It costs a ceiling about 6 points lower, which is a trade-off, not a free win.
- **The two-loop structure.** Same accuracy as the flat variant at one third of the $O(N^2)$ transport builds — and it is what makes the theorems apply to the code rather than to an idealisation of it.

5.2 What did not: the accuracy claim

The main table showed TRAIL 19–22 points ahead at a matched *budget*. The ablations then showed that margin is carried entirely by the control-conditioned relative-position bias inside the transport module — removing it costs 20.9 points — and **no baseline had that module**. The comparison was matched in belief updates and unmatched in inductive bias. Giving every baseline the same module (identical +23,012 parameters) leaves mirror descent as the only difference:

Model	No prior	With prior	Change	Belief updates
Single cross-attention	0.619	0.896	+0.277	1
MAC	0.609	0.905	+0.296	12
Free-form latent recursion	0.602	0.721	+0.119	12
TRAIL (ours)	0.812	0.812	—	4.32

Reversed, not narrowed

MAC with a matched prior beats TRAIL by 9.3 points, and a *single* relational attention step beats it by 8.4 while spending one belief update against twelve. By Prop. 1 attention is one TRAIL step, so the T=1 case of the method outperforms the method. **And the benchmark is the deeper problem:** if one dense transport application reaches 89.6%, HOPWORLD does not require iterated reasoning and cannot test the hypothesis this work is about. The benchmark is ours, so that is a design error, not a property of an inherited task.

The measured prices belong beside the benefits, not in a footnote: the prescribed step size costs **7.7 points** against a learned one (which converges to $\eta = 0.961$, violating Thm. B's $\eta \leq 0.25$), and the simplex constraint costs **8.4 points** against one unconstrained relational attention step.

6. The negative result

Proposition 3 predicts that halting time scales as $1/\Delta$ in the decision margin, so harder questions should receive more steps for free. **On HOPWORLD this does not happen.** Sweeping the outer threshold δ on a trained TRAIL model:

δ	mean hops	std. dev.	$\rho(\text{halt}, \text{true hops})$	accuracy
0.5	2.00	0.000	undefined	73.3%
0.3	2.00	0.000	undefined	73.3%
0.2	2.00	0.000	undefined	73.3%
0.1	2.00	0.000	undefined	73.3%
0.05	2.00	0.026	0.011	73.3%
0.02	2.21	0.408	0.057	76.7%
0.01	2.64	0.603	0.076	79.3%
0 (fixed depth)	4.00	0.000	undefined	81.2%

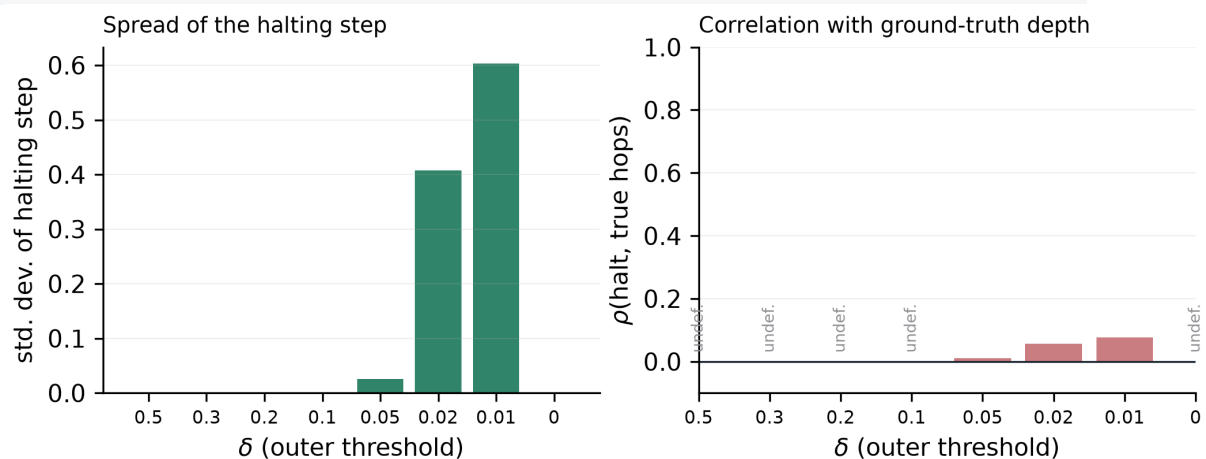


Figure 5. Left: over most of the threshold range the outer rule saturates — every example halts at exactly hop 2. Right: the correlation is therefore undefined for lack of variance, not measurably zero; where variance exists it is 0.06–0.08.

Two things must be separated here. Over most of the range the rule **saturates**: the standard deviation of the halting step is 0.000, so ρ is undefined rather than zero. Where there is variance (δ

$= 0.01-0.02$), $p = 0.06-0.08$, i.e. nothing. Reporting p at a single threshold would have hidden which of these was happening — a saturated rule and a genuinely uncorrelated one look identical through one number — which is why the evaluation now reports the spread alongside the correlation at every threshold.

Diagnosis

Prop. 3 is a statement about iterating a *fixed* energy, and it is true there — the test verifies the bound. The gap between theory and measurement is that M_h is a **dense learned operator**, not a single ground-truth relation: one application can compose several hops of the underlying task, so the number of TRAIL hops has no reason to equal the number of program hops. The theory bounds how long it takes to solve a hop; it says nothing about how many of the task's hops the model packs into one.

This is a testable diagnosis rather than a shrug, and it names the follow-up: constrain M_h (low rank, spatial locality, or a penalty on composing) so that one application is one relation, then re-measure p . Until that is done the contributions claimed are compute adaptivity and interpretability, **not** unsupervised difficulty estimation.

7. Three corrections made during development

Recorded because each changed a number, and because a reader running the artefact would otherwise rediscover them.

What was wrong	Why it mattered	Fix
The energy was re-conditioned at every step, so no step optimised a fixed objective.	The theorems did not apply to the code, only to an idealisation of it.	Two loops: hops outside, mirror descent on a fixed E_h inside. A test now pulls (s_h, M_h) out of the module and replays the inner loop through the certificate functions.
The Frank-Wolfe gap decided how many <i>hops</i> to take.	Cost ~ 16 accuracy points: it stopped early on exactly the multi-hop questions adaptivity is supposed to help.	Gap governs iterations within a hop; a belief-stationarity rule governs the number of hops.
<code>evaluate()</code> defaulted its depth to <code>cfg.n_steps</code> — TRAIL's <i>hop</i> count.	Baselines were trained at 12 iterations and evaluated at 4. The compute axis was wrong by $3\times$ and the comparison flattered TRAIL.	Depth now comes from each reasoner's own configuration; <code>test_eval_budget.py</code> asserts it and that budgets match.

On the third one

The affected runs were discarded and the comparison re-run. Re-measured, free-form latent recursion scores 60.2% at its true 12 updates — its accuracy was unchanged, because the eight iterations it was denied were adding nothing. The bug mattered for the *compute axis*, not the accuracy axis, and the qualitative conclusion survived. That was not knowable before re-running it.

8. Running it

Kaggle. Open `notebooks/trail_wacv2027_kaggle.ipynb`, enable the GPU accelerator and Internet, attach a CLEVR v1.0 dataset, run top to bottom. It starts in smoke mode (minutes end to end); set `SMOKE` and `QUICK` to `False` for the real run.

```
python tests/test_theory.py # the maths, ~1 min
python tests/test_eval_budget.py # matched budgets
python scripts/run_synthetic.py --quick # controlled study
python scripts/run_clevr.py --split iid # the real run
python scripts/run_clevr.py --split depth # depth extrapolation
python scripts/run_gqa.py # real images
python scripts/report.py --runs runs/synthetic # tables, MD + LaTeX
```

Stage	Cost on one T4
Theory + budget tests	~1 min (CPU)
HOPWORLD, all 5 models	~20 min
Featurising 20k+5k CLEVR images (once, shared by every model)	~25 min
Training one reasoner on CLEVR	~35–50 min
Five models + two generalisation splits	fits a 12 h session

9. Limitations

- The certificate bounds the hop's energy, not answer correctness.
- Only the *visual* state is drift-free; the control unit is an unconstrained recurrence. Extending the guarantee to the control path is open.
- M_h is $O(N^2)$ per hop. Fine at $N = 196$, and built once per hop rather than once per update, but it is the scaling bottleneck for finer atom grids. Low-rank or sparse transport is the obvious next step — and, per §6, may also be what makes halting track difficulty.
- Convexity holds for the relational penalty as written; a concave coherence reward would lose Thm. A's global guarantee and leave only stationarity.
- Everything here is a from-scratch reasoner over frozen features. Whether the same parameterisation helps inside a pretrained MLLM's latent reasoning loop is untested and is the natural follow-up.
- Dataset slugs could not be verified from the authoring environment; the loaders discover paths instead, and the CLEVR reader is tested against a deliberately hostile fixture, but the Kaggle attachment step needs a human eye once.

10. What a reviewer will press on

Likely objection	Prepared answer
"Attention already does this."	Prop. 1: attention is literally the $T=1$ case. The paper's question is what iterating it under a certificate buys, and <code>attn1</code> is the control that isolates exactly that.
"Deep equilibrium models / unrolled optimisation are not new."	Agreed, and cited. What is new is that the optimisation variable is the reasoning state itself, constrained to the hull of the image's own atoms, and that the stopping rule is the algorithm's own certificate rather than a learned gate.
"The accuracy is far below published CLEVR numbers."	Stated up front (§4). Different features, different budget; the comparison is internal and every model shares one cache.
"Does this actually beat anything?"	No, and we say so in §5.2. Given a matched prior, MAC and a single attention step both exceed it. The contribution is the certificate, the on-manifold state and the trace, together with an honest account of what they cost.

“Your halting does not do what you said.”	Reported as a negative in §6, with the saturation analysis and a mechanistic diagnosis, before a reviewer can find it.
“Is any of this reproducible?”	tests/test_theory.py checks all ten formal claims in about a minute; the notebook runs the study end to end.

Generated from the live run artefacts. Regenerate with `python report/make_report.py` after further models finish, and the tables update in place.